Data Structures in R

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Data School
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Recap

- What is R? And why learn R?
- Familiarity with the RStudio interface (opened a file)
- Variable assignment and computations in R
- Basic data types in R

What we will cover today

- Data Structures in R
- Missing values in R

Functions and Data Objects

- A function is a chunk of codes that conducts a specific task. R comes with many built in functions (e.g. mean, summary, t.test, cor.test, and so on)

```
Use help to look up these functions:
    <-
    read.csv
    write.csv
    plot
    print
```

Functions and Data Objects

- Data comes in many types recognized by R and RStudio
- The most common types are:

Data Type	Also Known as	Example
Numeric	Float	10, 2.1, -0.9
Character	String or text	“Calgary”, “Ottawa”, “Intervention”, “Control”
Logical	Boolean	TRUE, FALSE

Data Structures in R

Data Structures in R: Vectors

- A **vector** is a sequence/collection of data elements of the same basic type
- Members in a vector are officially called **components.**
- To assign multiple values to a variable, we can use an R object called a '**vector**'
- To create a vector, you should use **c()** function which means to combine the elements into a vector. **For Example:** *vect = c(14,26,38,30)* Position 3

  1 2 3 4
- To access a specific element in the vector, we simply need to call *variable_name[i]*, *'i'* being the element's position in the vector. **For example:** *vect[3]* would return *38* Vect[3] . Vect[1]

Source: Slides from Great Learning Academy on Introduction to R

Data Structures in R: Matrices

- A matrix is a sequence/collection of data elements of the same basic type arranged in a two-dimensional rectangular layout.
- A matrix is created using the **matrix()** function

```
my_matrix <- matrix(svect1, nrow=5, ncol=2)
my_matrix
```

- Being a 2-dimensional object, in order to obtain a specific value within the matrix, 2 coordinates needs to be entered.

Source: Slides from Great Learning Academy on Introduction to R

Data Structures in R: Matrices

vect[i] row x column

my_matrix[i,j] would return the element on the *ith* row, in the *jth* column

my_matrix[i,] would return the entire *ith* row

my_matrix[, j] would return the entire *jth* column

mat1 =

Extract rows and columns (vectors), and cells from **matrices** using square brackets [row, col].

- For example, if you would like to extract all the values of the 3rd row in *my_matrix*, you can use *my_matrix[3,]*. mat1[,2] mat1[5,]

  `Extract from my_matrix 1) the 2nd column, 2) the 5th row, and 3) cell that is in the 3rd row and 1st column`

  my_matrix[,2]; my_matrix[5,]; my_matrix[3,1]

row column! mat1[,1] mat1[3,2]

mat1[2,3]

Data Structures in R: Dataframes

- A data frame is used for storing data tables.
- It is a list of vectors of equal length. Unlike matrices, it can gather vectors containing **different** basic types
- Selection of specific elements in data frames works the same way as for matrices.
    - **For example:** *my_dataframe[i,j]* would return the element on the *ith* row, in the *jth* column

Source: Slides from Great Learning Academy on Introduction to R

Lists

- A list in R allows you to gather a variety of objects under one name (that is, the name of the list) in an ordered way.
- These objects can be matrices, vectors, data frames, even other lists, etc. It is not even required that these objects are related to each other.
- To create a list on R, use the **list()** function.
- To access the *ith* object in the list, write *my_list*[[i]]
- If you want to access a variable in the *ith* object in the list, write *my_list*[[i]][**variable coordinates**].

Source: Slides from Great Learning Academy on Introduction to R

Missing Values in R

Missing values are denoted by NA or NaN for undefined mathematical operations.

- is.na() is used to test objects if they are NA
- is.nan() is used to test for NaN
- A NaN value is also NA but the converse is not true

Source: https://www.coursera.org/learn/r-programming/lecture/HW3TB/data-types-missing-values

Other Resources

- https://swcarpentry.github.io/r-novice-inflammation/13-supp-data-structures/
- https://www.geeksforgeeks.org/data-structures-in-r-programming/
- https://data-flair.training/blogs/data-structures-in-r/
- https://warwick.ac.uk/fac/sci/moac/degrees/moac/ch923/r_introduction/help/
- https://towardsdatascience.com/dealing-with-missing-data-in-data-science-projects-e8ac7a4efdff
- https://jameshoward.us/2016/07/18/nan-versus-na-r/
- https://www.r-bloggers.com/2012/08/difference-between-na-and-nan-in-r/
- **Important:** https://cran.r-project.org/doc/manuals/r-release/R-lang.pdf

to be continued...

thanks for listening.


Thank You